MBAN5570

ACCOUNTING & FINANCIAL ANALYTICS

Assignment 3

Risk-Adjusted NPV Analysis and Cryptocurrency Time Series Modeling

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Part I: Pharmaceutical Drug Development

Monte Carlo NPV Analysis

This analysis evaluates the risk-adjusted Net Present Value (NPV) of developing the NIAGARA drug using a 5,000-iteration Monte Carlo simulation. The project proceeds through four sequential stages: Research and Development (R&D), Pre-Clinical Testing, Clinical Trial Phase I, and Clinical Trial Phase II - with uncertain cost, duration, and success probability at each stage. A 15% discount rate was applied, stage costs were discounted at the start of each phase, and market profit was realized only if all four development stages were completed successfully.

NPV Statistical Summary	Value
Mean NPV	$60.49M
Median	$-80.62M
5th percentile	$-290.11M
25th percentile	$-96.42M
50th percentile	$-80.62M
75th percentile	$-69.11M
95th percentile	$1,230.81M
Probability NPV > 0	6.28%

The expected NPV is $60.49M, while the median NPV is -$80.62M. This indicates a strongly right-skewed distribution: most simulated outcomes are unprofitable, but a small number of very large successful outcomes raise the average above zero. The probability of a positive NPV is 6.28%, which indicates that profitable outcomes are relatively rare under the simulated assumptions.

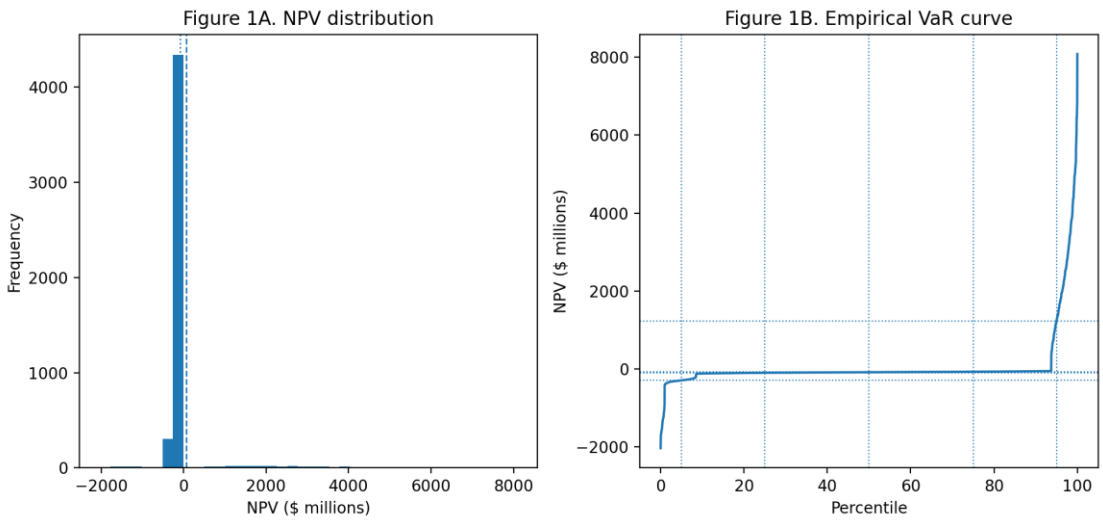

Figure 1. Empirical NPV distribution and Value-at-Risk (VaR) view for the NIAGARA project.

Figure 1 shows that the empirical distribution is concentrated in negative outcomes, with a long positive tail generated by a small number of highly successful simulations. This distributional pattern is consistent with pharmaceutical development, where losses are common but occasional large payoffs can create positive expected value.

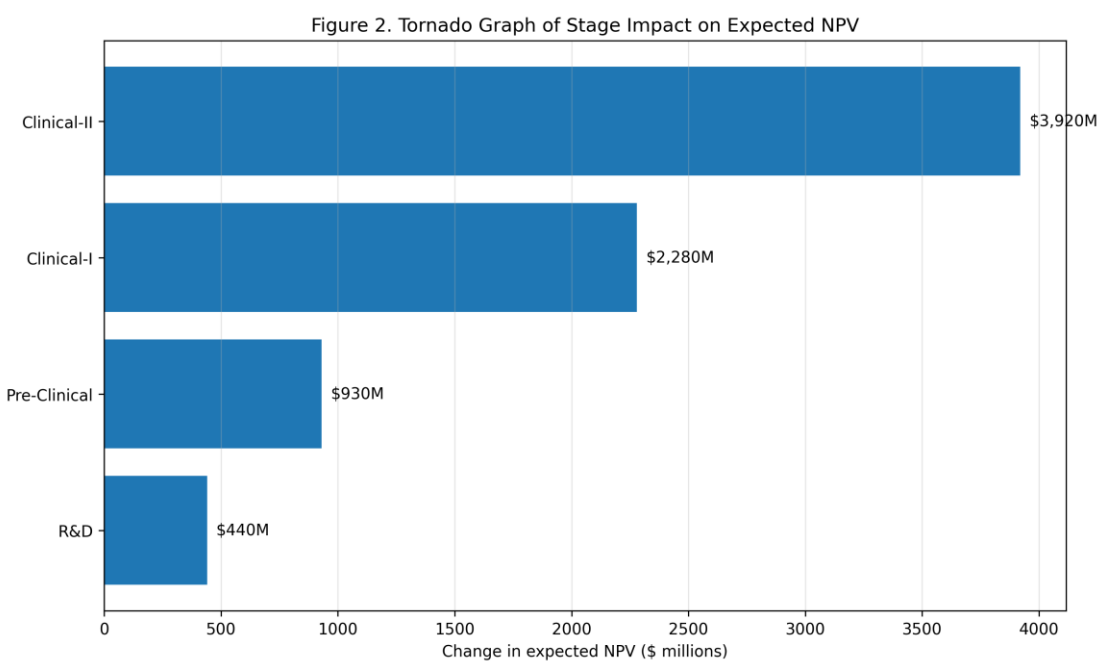

Figure 2. Tornado-style stage impact on expected NPV.

Clinical Phase II is the most critical stage-related driver of project value. Changing Clinical Phase II from failure to success produces the largest increase in expected NPV, followed by Clinical Phase I, Pre-Clinical, and R&D. This result is economically intuitive because substantial investment has already been committed by Clinical Phase II and the project is closest to commercialization.

Table 2. Stage-by-stage success rates and NPV impact analysis.

Stage	Success Rate	E[NPV | Fail]	NPV Impact ($M)
R&D	32.28%	$-80.18	$435.78
Pre-Clinical	45.48%	$-89.94	$979.71
Clinical-I	49.86%	$-293.85	$2,373.70
Clinical-II	85.25%	$-1,371.57	$4,048.78

Among the development stages, Clinical Phase II has the largest NPV impact at $4,048.78M. This makes it the most important decision gate in the NIAGARA project. Based on expected value, the project appears acceptable; however, the stand-alone risk remains high, so the investment is more justifiable within a diversified pharmaceutical development portfolio than as an isolated low-risk opportunity.

Strategic Priorities

1. Market potential assessment should receive primary analytical focus because final market profit is the strongest overall driver of NPV.
2. Clinical Phase II timeline and protocol design should be prioritized because delays at this stage have the largest adverse effect on project value.
3. Stage-gate discipline remains important so that low-potential projects can be terminated promptly and capital can be reallocated.
4. Early-stage costs matter, but their effect on NPV is materially smaller than the effect of late-stage execution and commercialization potential.

Part II: Bitcoin Time Series Analysis

2.1 Data Preparation and Exploratory Analysis

After arranging the data in chronological order, the closing price, log price, daily log return, and daily log volume growth series were constructed. The closing price series does not fluctuate around a stable mean and therefore does not appear stationary. In contrast, the return series is much closer to stationary, although its variance changes over time. The plots also show distinct periods of calm and periods of large swings, indicating that volatility is not constant.

The Augmented Dickey-Fuller (ADF) test supports this visual evidence. For the closing price series, the test statistic is -1.1839 with $p = 0.6805$, so the null hypothesis of a unit root cannot be rejected. For the return series, the test statistics are -7.8891 with $p < 0.001$, indicating that the return series is stationary.

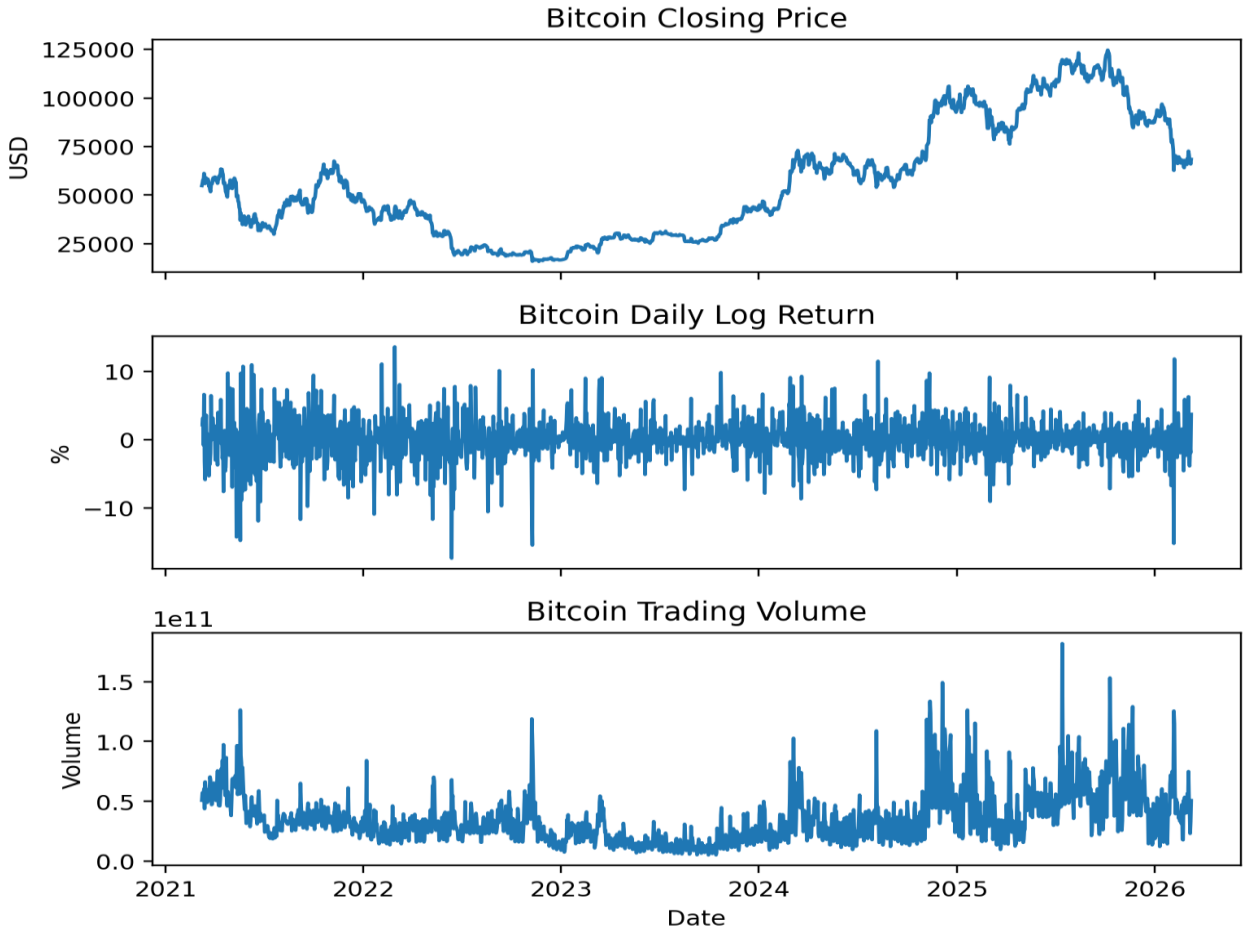

Figure 3. Bitcoin closing price, daily log return, and trading volume.

2.2 Moving Average and Exponential Smoothing Forecasts

To ensure consistency with the aligned sample used for returns, the forecasting analysis was based on 1,826 observations, with the last 366 days reserved as the test set. Table 3 compares the performance of MA(5), MA(20), SES (α = 0.2), and SES (α = 0.8) using CFE, MSE, MAD, and MAPE.

Table 3. Comparative forecast accuracy metrics (test set, n = 366).

Method	CFE	MSE	MAD	MAPE
MA(5)	-60,204	9,037,072	2,232.68	2.34%
MA(20)	-221,247	29,384,954	4,148.39	4.44%
SES (α=0.2)	-101,023	11,445,427	2,524.96	2.67%
SES (α=0.8)	-22,975	4,595,771	1,563.27	1.66%

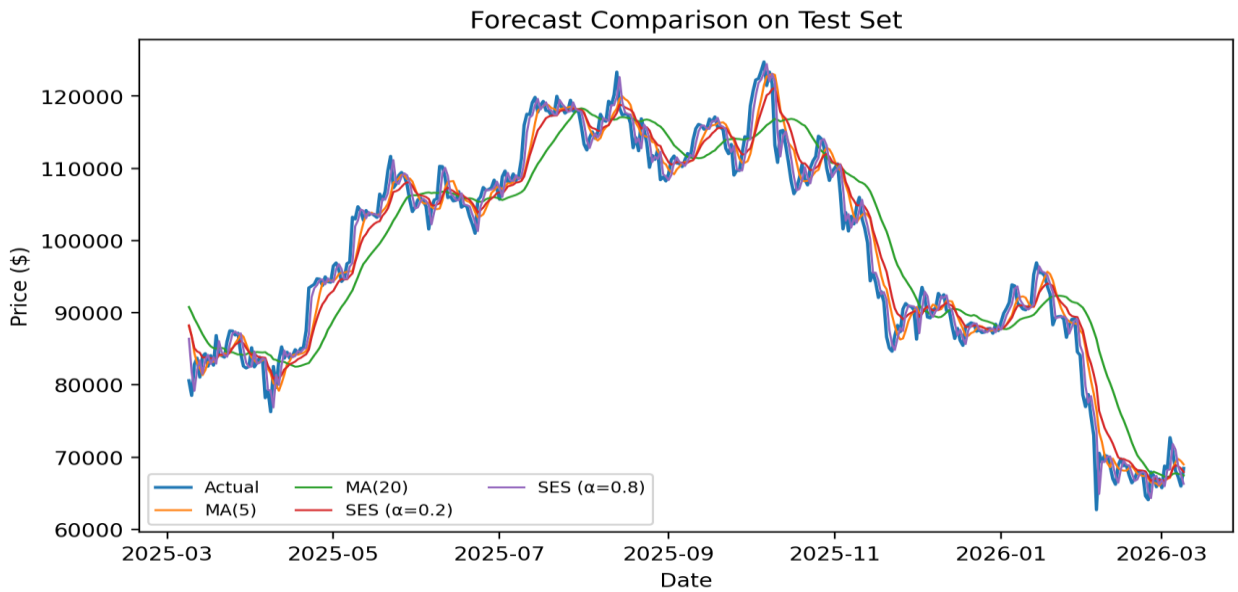

Figure 4. Out-of-sample forecast comparison on the test set.

SES (α = 0.8) performs best across all four error measures. Relative to MA(5), its MSE is approximately 49% lower; relative to MA(20), the reduction is about 84%. This suggests that Bitcoin prices respond quickly to new information and that a method placing greater weight on the most recent observation is better able to adapt to rapid market changes.

2.3 Autocorrelation and Stationarity

The ACF and PACF of daily log returns were examined up to lag 20. Most autocorrelations lie within the approximate 95% confidence bounds of ±0.0459, with only four notable spikes at lags 1, 8, 9, and 13. There is no sustained cutoff or decay pattern that would strongly support a simple AR, MA, or ARMA specification.

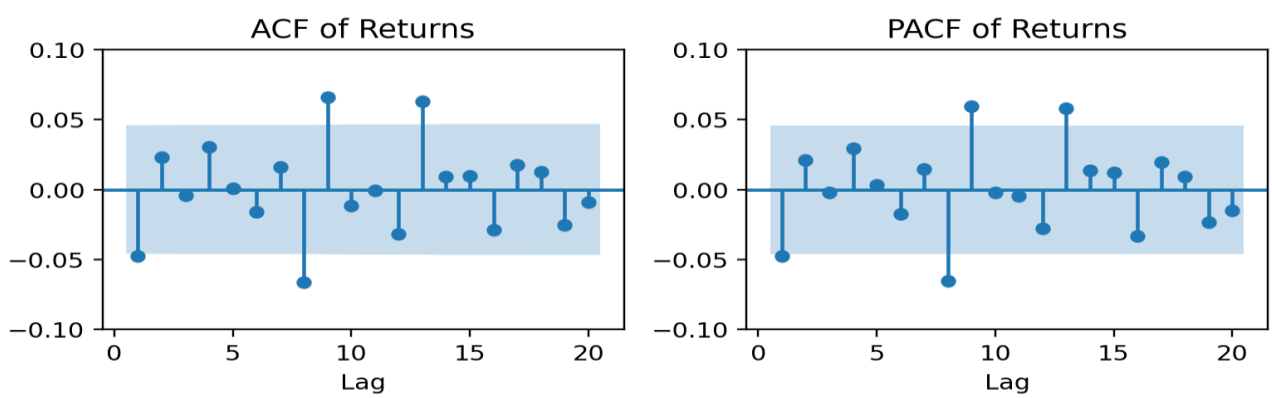

Figure 5. ACF and PACF of daily Bitcoin returns up to lag 20.

Overall, the return process is closer to weak white noise than to a stable AR, MA, or ARMA process. Financial prices are usually non-stationary because new information can permanently change the price level. Returns are closer to stationary because differencing removes the unit root in prices, leaving a series that fluctuates around a relatively stable mean.

2.4 ARIMA Forecasting

An AR(1) model was estimated for returns, while ARIMA(1,1,0) and ARIMA(0,1,1) models were estimated for log prices for comparison. The ARIMA price models slightly outperform the best smoothing model, although the improvement is very small in practical terms.

Table 4. ARIMA estimation and out-of-sample comparison.

Model	Key coefficient	p-value	AIC	Test MSE
AR(1) returns	-0.0393	0.058	7440.93	-
ARIMA(1,1,0) log price	-0.0396	0.054	-6003.43	4,493,165
ARIMA(0,1,1) log price	-0.0384	0.060	-6003.33	4,493,256

The estimated AR and MA coefficients are small and only marginally significant. This implies that past price movements provide very limited predictive information for future Bitcoin returns, even though ARIMA models generate a slight reduction in forecast error relative to SES ($\alpha = 0.8$).

2.5 Volume and Volatility Clustering

Squared returns were used as the volatility proxy, defined as $VP_t = r^2_t$. Squared returns display clear clustering, with periods of high volatility tending to occur together and periods of low volatility also appearing in clusters. This pattern is inconsistent with the assumption of constant conditional variance used in classical linear models and instead suggests the presence of conditional heteroskedasticity.

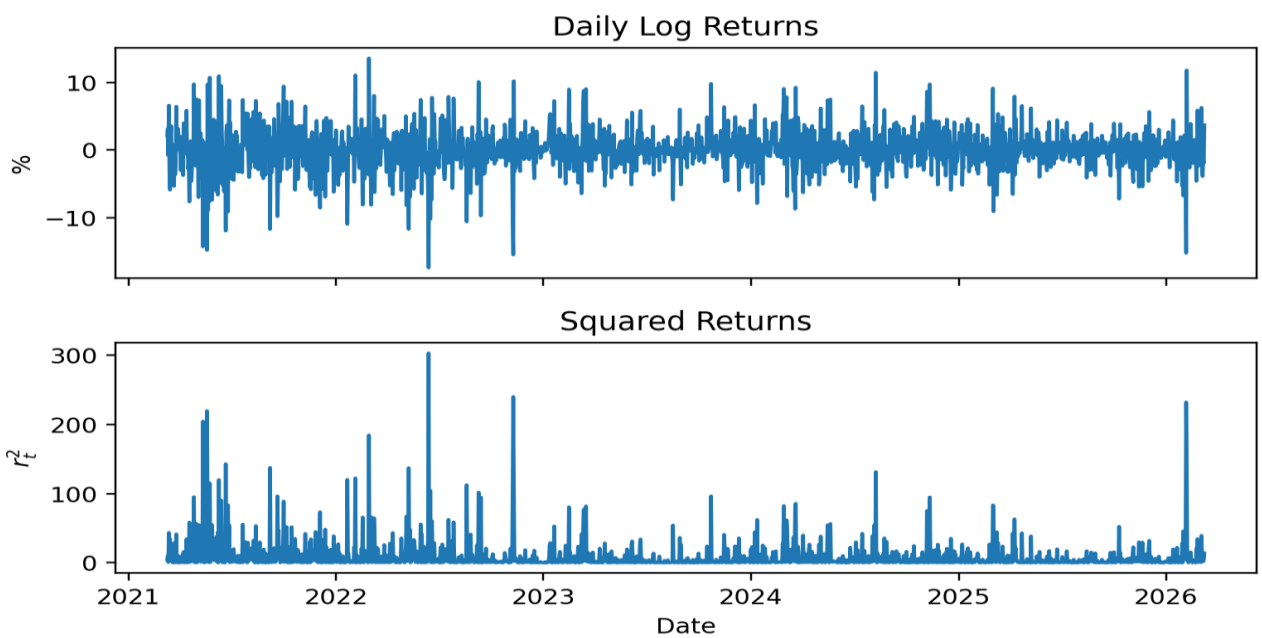

Figure 6. Daily log returns and squared returns.

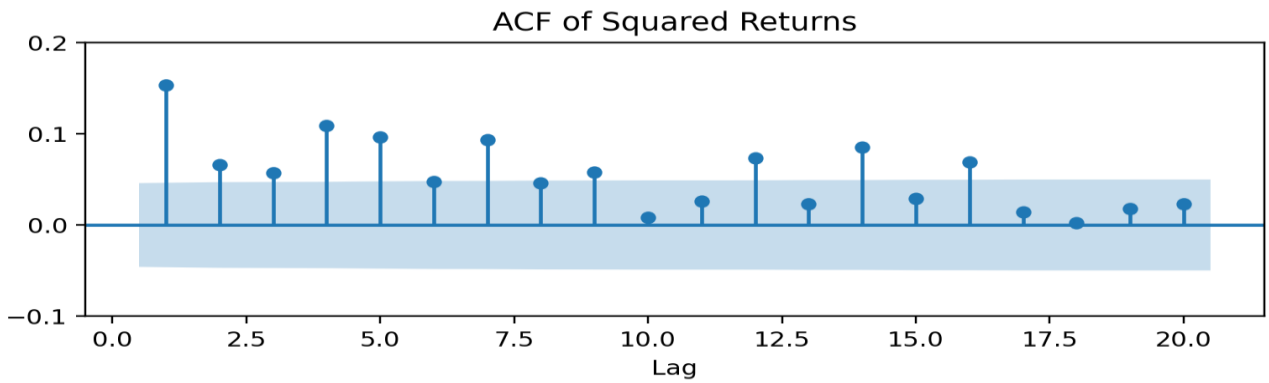

Figure 7. ACF of squared returns.

The ACF of squared returns shows 12 significant lags out of the first 20, which is much stronger than the autocorrelation pattern observed in raw returns. This indicates that while the mean of returns is only weakly predictable, the variance of returns is persistent and time-varying.

Table 5. Volatility regressions.

Model	Key result	Significance	Fit
Squared-return AR(1)	$\gamma_1 = 0.1534$	$p < 0.001$	$R^2 = 0.024$
Extended volatility model	$\delta_1 = 0.1693$; $\delta_2 = 1.9156$	$p < 0.001$ for both	$R^2 = 0.105$

In the simple autoregressive volatility model, $\gamma_1 = 0.1534$ ($p < 0.001$), indicating that higher volatility yesterday is associated with higher volatility today. In the extended model, both daily log volume growth ($\delta_1 = 0.1693$) and lagged absolute return ($\delta_2 = 1.9156$) are positive and statistically significant. The explanatory power rises from 2.4% to 10.5%, showing that trading activity and recent market shocks provide useful additional information about current volatility.

A GARCH(1,1) specification is appropriate for this setting because the return series is close to white noise in the mean while the variance is clearly time-varying. A parsimonious specification is $r_t = \mu + \varepsilon_t$ for the mean equation and $\sigma^2_t = \omega + \alpha\varepsilon^2_{t-1} + \beta\sigma^2_{t-1}$ for the variance equation. Such a model is useful for Value-at-Risk estimation, portfolio risk management, and trading decisions that depend on changing volatility conditions.

Conclusion

The NIAGARA project has a positive expected NPV but a high probability of loss, so it is better viewed as a portfolio-style investment than as a stand-alone low-risk project. For Bitcoin, the analysis indicates non-stationary prices, stationary returns, limited return predictability, and strong volatility clustering. Taken together, the assignment shows that expected value, forecast accuracy, and volatility dynamics can lead to very different but complementary business conclusions from the same underlying data.